

Ethereum dApp Final Project Ideas

A. Decentralized Finance (DeFi) & Insurance

1. **Uniswap-Style DEX (Automated Market Maker)** – A simple decentralized exchange where users can swap ERC20 tokens via liquidity pools github.com.
2. **Collateralized Lending/Borrowing Platform** – A mini Compound-like dApp for lending ERC20 tokens. Users supply tokens and borrow others against collateral github.com.
3. **Yield Farming & Staking DApp** – Stake an ERC20 token to earn reward tokens over time github.com. For example, users lock a stablecoin and receive a project token as yield.
4. **Token Crowdsale (ICO) Platform** – An Initial Coin Offering dApp where a new ERC20 token is sold for ETH. OpenZeppelin's sample crowdsale contract can be used github.com.
5. **Simple Lottery DApp** – A straightforward lottery where users buy tickets (pay an ERC20 or ETH) and a random winner takes the pot. For example, a contract where participants send ETH and one winner is chosen at end github.com.
6. **Flight Delay Insurance** – Passengers purchase “flight insurance” by paying an ERC20 premium. If their flight is delayed (data provided by an oracle), the contract pays out a compensation token or ETH github.com.
7. **Continuous Token Stream (Salary Streaming)** – A payment dApp where money streams over time (e.g. paying wages per second). Similar to Sablier, a contract holds ERC20 tokens and continuously releases them to the payee.

B. NFT Platforms & Marketplaces

1. **NFT Marketplace (OpenSea-Lite)** – A basic marketplace for NFTs where users can list ERC721 tokens for sale and others can buy github.com. Includes listing creation, purchase, and perhaps a simple fee to the marketplace.
2. **NFT Auction House** – An auction dApp where users can trustlessly auction their ERC721 NFTs to the highest bidder within a time frame github.com. Sellers specify an ERC20 token as the bid currency (could be ETH, DAI, etc.).
3. **Music NFT Platform** – A specialized marketplace for musicians to mint their songs or albums as NFTs. Fans can buy and resell these music NFTs, and perhaps play the music in-app github.com.
4. **Event Ticketing DApp** – Issue event tickets as ERC721 NFTs instead of paper tickets. Organizers mint NFT tickets for an event; users buy them, and ownership is verified at venue entry (scan QR code tied to NFT) github.com.

C. Blockchain Games & Collectibles

1. **Decentralized Tic-Tac-Toe** – A two-player tic-tac-toe game entirely on Ethereum. A player can create a game, another joins, and they alternate moves stored in the contract. To make it interesting, players stake a small amount of ERC20, and the winner takes the pot. *Smart Contracts*: Game contract tracking the board state and bets (could use ERC20 or just ETH). *Repo*: Tic-Tac-Toe DApp (GitHub) – demonstrates using tokens as bets and locking them until game end github.com.
2. **Rock-Paper-Scissors with Tokens** – A simple two-player game where each player commits to rock, paper, or scissors off-chain (e.g., by hashing their move), then reveals it. The winner earns an ERC20 reward (or wins the staked tokens from the loser) in a trustless way github.com.
3. **Coin Flip Betting Game** – A multiplayer coin-flip game: either two players bet against each other on coin toss, or a player bets against the house contract. The result is determined by a fair random source. If heads, player1 (or house) wins; if tails, player2 wins. *Smart Contracts*: A coin flip contract that accepts bets in ERC20 (or ETH), requires two equal bets, and on flip (e.g., using VRF) transfers the pot to the winner github.com.

4. **On-Chain Chess Game** – A fully decentralized chess game where the moves are recorded on-chain and enforceable by the contract. Two players join a match and alternate moves. If a player fails to move within a time limit, they lose by forfeit. github.com.

D. Supply Chain & Provenance

1. **Farm-to-Table Supply Chain (Coffee)** – Track an agricultural product (e.g., coffee beans) from farm to consumer using NFTs or tracked events on Ethereum. Each batch is represented by a unique token or ID that gets updated at each step (harvest, export, roast, etc.) github.com
2. **Pharmaceutical Supply Chain (Drug Traceability)** – Ensure authenticity of medicines by tracking them on Ethereum. Each drug package is an NFT or recorded in a contract from the manufacturer, through distributors, to pharmacies github.com. Authorized parties update the state (produced, shipped, received) to fight counterfeits.
3. **Product Provenance Verification** – A generic supply chain template for any product (fashion, electronics, art). The dApp registers products as NFTs and records each ownership or custody change. Consumers can verify an item's origin and history. *Smart Contracts*: ERC721 for each item (or ERC20 for commodity bulk tracking) and functions to update status/location. *Repo*: Sample Supply Chain DApp (GitHub) – allows producers to add products and customers to track orders github.com.

E. Healthcare & Medical

1. **Electronic Health Records (EHR) Management** – A prototype for managing patient records on Ethereum with privacy. Doctors and patients have addresses; a patient's records (encrypted off-chain, e.g. IPFS) are referenced on-chain for integrity github.com. Only authorized addresses can add or view a patient's record hash.
2. **COVID-19 Vaccination Certificates (NFT)** – A dApp for issuing and verifying vaccination certificates as NFTs. A health authority mints a non-fungible token to a patient's address as proof of vaccination (could embed a hash of the actual certificate PDF). Verification can be done by checking the NFT on-chain (no personal data exposed) github.com

3. **Organ Donation & Transplant Platform** – A blockchain platform to transparently connect organ donors with recipients, ensuring fairness in allocation. Donors can register their intent (perhaps as an NFT certificate), and when an organ becomes available, the system (with hospital input) matches to a recipient. github.com.

F. Social, Identity & Certification

1. **Decentralized Social Network (Microblogging)** – A Twitter-like dApp (e.g., Peepeth) where users post messages to the blockchain instead of a centralized server. Each post could be an event stored by a smart contract, or an NFT (for example, each “peep” as an NFT). Because writing all text on-chain is costly, one approach is to store IPFS hashes of posts on-chain. github.com.
2. **Charity Donation Tracker** – A platform connecting donors and charities with transparency. Charities are listed with Ethereum addresses; donors send funds (ETH or ERC20 stablecoin) through the dApp, which records each donation on-chain tagged with the project or cause github.com. Donors could receive an NFT receipt or a “thank you” token for their contribution.
3. **Certificate/Diploma Issuance Platform** – A dApp for institutions (like universities or online course providers) to issue graduation certificates or skill certifications as NFTs. Each certificate NFT contains metadata about the achievement and is assigned to the student’s wallet address, serving as a permanent, verifiable credential github.com.
4. **Document Timestamping Notary** – A proof-of-existence dApp where users can upload a document’s hash to the blockchain to get a timestamped proof. This can serve as a decentralized notary service to prove a document or data existed at or before the time of the transaction. For example, a writer could timestamp a manuscript draft to prove authorship. github.com.

G. Governance & Security Tools

1. **DAO Governance & Voting System** – A simple DAO platform where a governance ERC20 token is used to vote on proposals. Token holders can create proposals (e.g., text proposals stored via IPFS hash) and vote Yes/No. After a defined voting period, the proposal is executed if quorum and majority are met. This can be modeled on Compound or OpenZeppelin Governor contracts. github.com.
2. **Multi-Signature Wallet** – A contract wallet that requires M-of-N owners to approve a transaction before it executes. This is a vital security tool for managing funds in a team. The classic Gnosis Safe or parity multisig can be an inspiration. Owners can submit transactions (destination address + data + value) to the contract, and a certain number of other owners must confirm before it can be executed github.com.
3. **Token Vesting Contracts** – A dApp to lock and gradually release tokens for team members or investors. It can create vesting schedules (with a cliff time and vesting duration) for given beneficiaries. For example, Alice's 1000 ABC tokens vest over 1 year with monthly releases. The contract holds the tokens and allows Alice to claim the unlocked portion after each interval github.com.

H. Marketplaces & Commerce

1. **Decentralized E-Commerce Marketplace** – A platform like a blockchain eBay/Etsy. Sellers can list physical or digital products with a price (in ETH or stablecoin), and buyers can purchase directly. Upon purchase, payment is held by the contract until the item is marked delivered (could use a simple “confirm delivery” by buyer, or a time lock after which seller can withdraw if no dispute). github.com.
2. **Freelance Jobs Marketplace** – A decentralized Upwork: clients post job listings and freelancers can bid. On hire, the client's payment is escrowed. github.com.